Introduction to Artificial Intelligence

Lab 4: Regression and Classification

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1 Task and Dataset Description

The objective of this laboratory is to build a regression model to predict housing prices using the California Housing dataset. Provided via scikit-learn (`fetch_california_housing`), the dataset represents aggregate data from the 1990 California census. The target variable is `MedHouseVal`, which denotes the median house value for a specific block group, expressed in hundreds of thousands of dollars.

The dataset consists of 20,640 observations and eight numerical features: `MedInc`, `HouseAge`, `AveRooms`, `AveBedrms`, `Population`, `AveOccup`, `Latitude`, and `Longitude`. A critical anomaly in this dataset is that the target variable is artificially capped at 5.0 (representing $500,000). Any property valued above this threshold was recorded simply as 5.0, creating a strict upper bound that models must navigate. The data was split into training (80%) and testing (20%) sets, yielding 16,512 training instances and 4,128 test instances.

2 Data Preparation and Analysis

The initial phase of data preparation focused on feature analysis and identifying potential redundancies. A correlation matrix was computed to detect multicollinearity among the variables. While certain expected correlations were observed—most notably the strong positive relationship between `AveRooms` and `AveBedrms`—no features were found to be strictly linearly dependent. Instead of discarding one of the room-related variables, both were deliberately retained. The interplay between total rooms and bedrooms captures the internal layout and density of the property, which serves as a valuable predictor of market value. Consequently, the complete set of eight features was preserved for the modeling phase.

For data preprocessing, a targeted approach to normalization was implemented. Rather than applying a blanket transformation to the entire dataset, feature scaling (`StandardScaler`) was integrated exclusively into the Ridge Regression pipeline. Linear models rely on optimization algorithms and regularization penalties that are highly sensitive to the magnitude of input variables. In contrast, tree-based ensembles (Random Forest and XGBoost) were trained on the raw, unscaled data. Since these algorithms operate by finding optimal feature-value thresholds to partition the data (e.g., "Is Population ¿ 1000?"), their internal logic is fundamentally invariant to monotonic transformations like scaling.

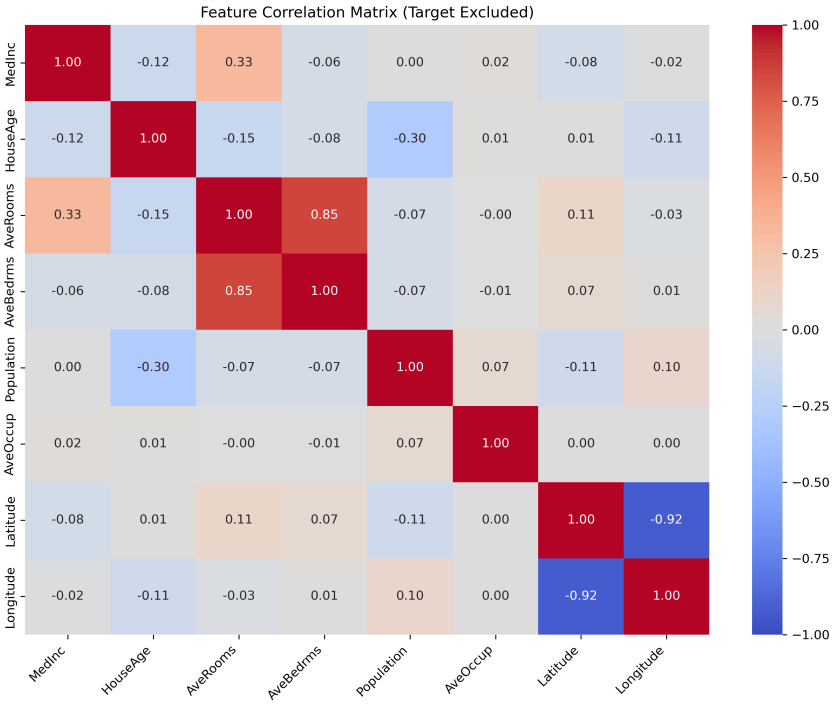


Figure 1: Feature Correlation Matrix.

3 Model Choice and Hyperparameters

To provide a comprehensive comparative analysis, three distinct architectures were selected. **Ridge Regression** serves as the linear baseline, applying an L_2 penalty to prevent overfitting, though its linearity limits its ability to capture complex spatial patterns like geographic coordinates. To address this, **Random Forest Regressor** was implemented as a non-linear *bagging* ensemble to capture complex feature interactions robustly. Finally, **XGBoost** was chosen as the state-of-the-art *boosting* benchmark, building sequential trees to systematically minimize residual errors.

For the hyperparameter tuning, we optimized each model's specific parameters using a Grid Search with 4-fold cross-validation, optimizing for the lowest Root Mean Squared Error (RMSE). For Ridge Regression, we tuned the regularization strength (`alpha`). For Random Forest, we optimized structural complexity through the total number of trees (`n_estimators`), the maximum tree depth (`max_depth`), and the minimum samples required to split nodes (`min_samples_split`). For XGBoost, we focused on learning dynamics and boosting architecture by tuning the number of boosting rounds (`n_estimators`), the maximum depth of the individual trees (`max_depth`), and the step size shrinkage (`learning_rate`).

Table 1: Hyperparameter grids explored during 4-fold CV and the optimal values selected.

Model	**Hyperparameters Tested**	**Optimal Value**
Ridge Regression	`alpha`: [0.1, 1.0, 10.0, 100.0]	0.1
Random Forest	`n_estimators`: [50, 100, 200]	200
	`max_depth`: [None, 10, 20]	None
	`min_samples_split`: [2, 5, 10]	2
XGBoost	`n_estimators`: [100, 200]	200
	`max_depth`: [3, 5, 7]	7
	`learning_rate`: [0.05, 0.1]	0.1

4 Results

Following cross-validation, the optimal estimators were evaluated on the unseen test set. As detailed in Table 2, XGBoost significantly outperformed both Random Forest and the baseline Ridge Regression, achieving the lowest test RMSE (0.4715) and the highest explained variance ($R^2 = 0.8250$).

Table 2: Final model comparison on the test set.

Model	Test MAE	Test RMSE	Test R^2
Ridge Regression	0.5332	0.7456	0.5758
Random Forest	0.3266	0.5039	0.8062
XGBoost	0.3120	0.4715	0.8250

Visual analysis of the predictions (Figure 2) reveals how each model handles the dataset's artificial 5.0 target cap. As a continuous mathematical plane, Ridge Regression frequently predicts values well above 5.0 for luxury homes. In contrast, the tree-based ensembles cannot physically extrapolate beyond the maximum observed training value, resulting in a strict vertical cutoff.

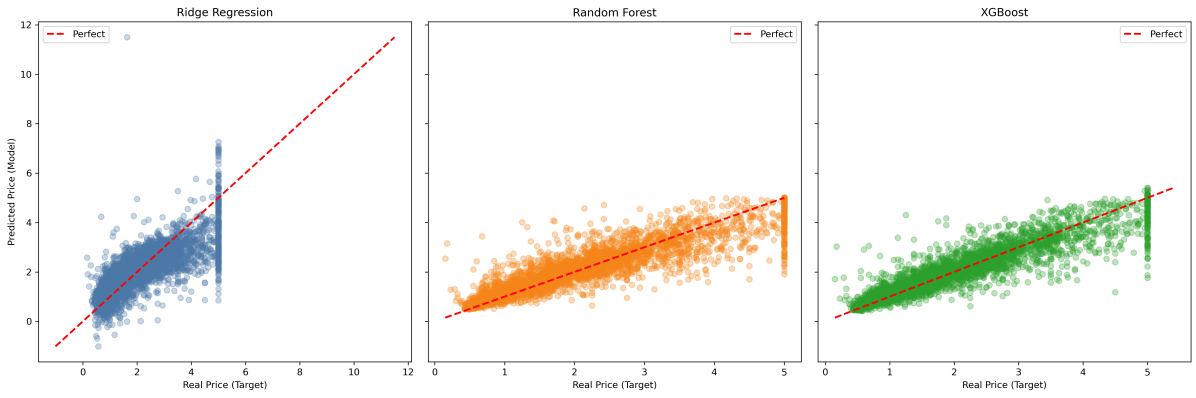


Figure 2: Actual vs Predicted values. Note the linear overshooting vs. the tree-based vertical cutoff at the 5.0 cap.

Furthermore, the feature importance extraction (Figure 3) highlights the internal logic of the models. While Ridge assigns weights based on scaled linear relationships, both Random Forest

and XGBoost overwhelmingly identify Median Income (`MedInc`) as the dominant predictor. Notably, XGBoost isolates this feature much more aggressively than Random Forest, assigning minimal importance to less relevant variables.

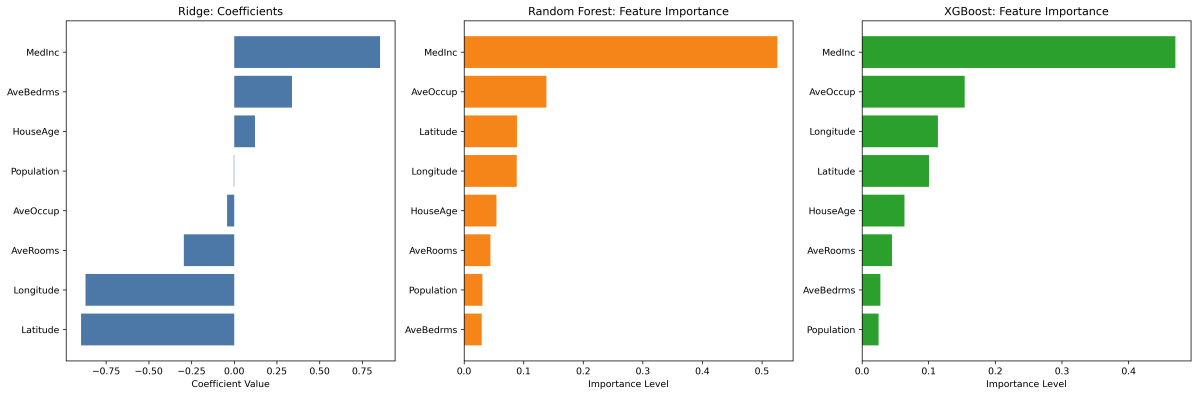


Figure 3: Feature Importance and Coefficients. The tree-based models heavily prioritize Median Income (`MedInc`).

5 Conclusion

In this experiment, predicting California housing prices revealed the severe limitations of linear models when dealing with non-linear spatial data and capped targets. Ridge Regression struggled to interpret raw geographic coordinates (Latitude/Longitude), whereas tree-based ensembles effectively segmented the spatial grid. The implementation of XGBoost yielded the best overall predictive performance. The distinct underestimation of high-end properties by all models near the 5.0 cap suggests that critical explanatory variables representing luxury prestige (e.g., proximity to the ocean) are missing from the original dataset.